Network

Architecture

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Arc's system architecture integrates the Malachite consensus layer with the Reth execution layer to process and finalize transactions.

Arc's architecture is composed of two core components: the **Consensus layer** and the **Execution layer**. Together, they provide deterministic finality, stable transaction fees, programmable privacy, and financial primitives purpose-built for stablecoin-native applications.

Consensus layer

Arc runs on **Malachite**, a high-performance implementation of the Tendermint Byzantine Fault Tolerant (BFT) protocol. Malachite ensures:

- **Deterministic finality:** Blocks finalize in less than one second.
- **Irreversibility:** Transactions cannot be reorganized or rolled back once committed.
- **Resilience:** Validators commit blocks under a Proof-of-Authority model.

The Consensus layer orders and finalizes transactions securely, providing institutional-grade guarantees for reliability and performance.

Execution layer

Arc's Execution layer is built on **Reth**, a Rust implementation of the Ethereum execution layer. It maintains the blockchain ledger and state, and extends it with components optimized for stablecoin finance:

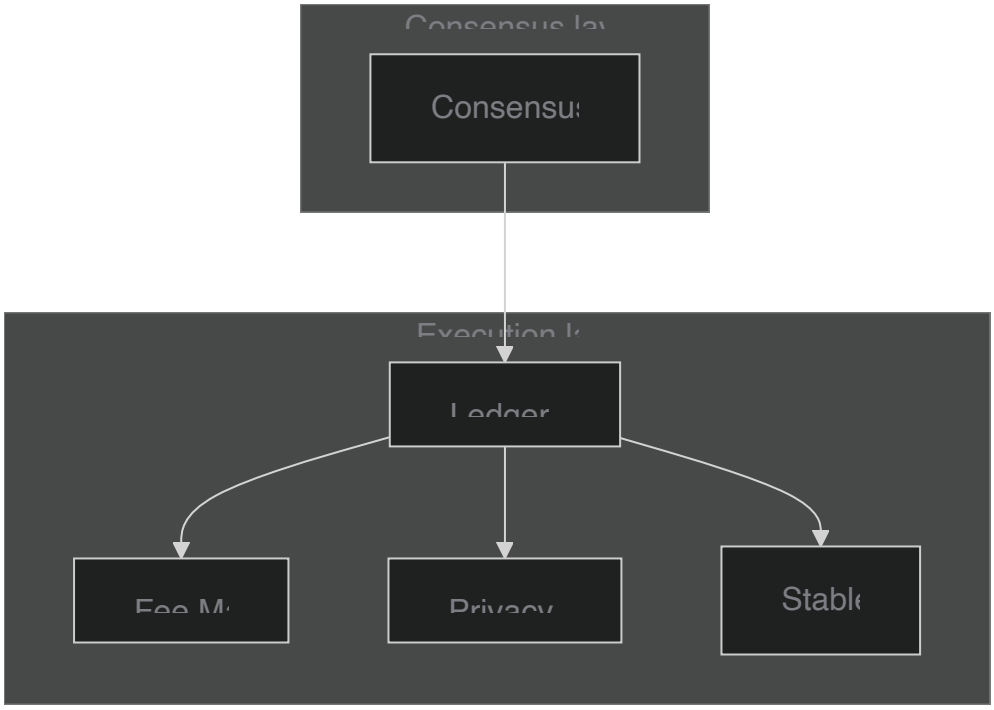
- **Ledger and State:** Stores accounts, balances, smart contracts, and transaction history.
- **Fee Manager:** Stabilizes and smooths fees using USDC as the unit of account.
- **Privacy Module:** Provides confidential transfers and selective disclosure via view keys.

- **Stablecoin Services:** Powers multi-currency payments, FX conversions, and programmatic settlement across supported stablecoins.

By combining these components, the Execution layer provides a familiar EVM-compatible environment with stablecoin-native extensions built in.

System architecture diagram

The diagram below shows Arc's architecture at a high level.



The **Consensus layer** determines the order of transactions and finalizes blocks. The **Execution layer** applies those transactions to the ledger and processes them through its internal modules.

Developer benefits

By understanding Arc's system architecture, you can:

- Trust that your transactions finalize instantly and irreversibly.
- Build on a familiar EVM-compatible stack (Reth) with stablecoin-native extensions.
- Leverage built-in financial components like Fee Manager, Privacy, **FX**, and paying in other stablecoins without needing external workarounds.

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